Supplementary Information for

A rising biocapacity total can mask ecological-capital land drawdown: a two-land model

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Equation numbers without the prefix S refer to the main text.

S1 Symbols and model extensions

A global hectare (gha) adjusts a hectare to world-average biological productivity for the relevant account year. Annual Ecological Footprint and biocapacity flows are reported in gha, with the annual basis implicit in that year (Global Footprint Network, 2009, 2019). In the model, B and E are annual account-equivalent quantities; in the empirical analysis, B and E_{prod} are observed NFA account values. Model contributions Y_f and Y_c and shortfall S use the same units, but are not separate NFA series. The debt state D is a leaky state driven by aggregate deficit $E - B$ and is measured in gha yr; $\dot{D}$ and ηD are in gha.

Table S1: Symbols, definitions, units, and roles. Per-hectare yields and contributions use the annual account basis (gha ha^{-1}). The coefficient $b_{G,c}$ is in gha yr ha^{-1}; annual account totals are in gha, with the accounting year implicit. The leaky debt state D is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	uncalibrated regeneration/readout state (quality index)	dimensionless	state
K_q	admissible quality set, $[0, q_{\max}]$	dimensionless	imposed interval
P	human population	cap	state
D	leaky-debt state driven by aggregate deficit $E - B$	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	OWID/FAOSTAT cereal yield, a physical productivity indicator (not all crops)	tonnes ha^{-1}	empirical input

Symbol	Meaning	Unit	Role
$\hat{b}_f(t)$	Base-anchored index-scaled proxy yield used only in SI S5	$\mathrm{gha\,ha^{-1}}$	proxy construction
b_{f0}	base fast-land yield per hectare	$\mathrm{gha\,ha^{-1}}$	parameter
b_c	capital-land direct yield	$\mathrm{gha\,ha^{-1}}$	parameter
$b_{G,c}$	capital regeneration-value coefficient	$\mathrm{gha\,yr\,ha^{-1}}$	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1-q/q_{\max})$	$\mathrm{yr^{-1}}$	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	$\mathrm{gha\,ha^{-1}}$	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	$\mathrm{gha\,ha^{-1}}$	derived
$Y_f,\ Y_c$	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived
B	model biocapacity $Y_f + Y_c$; or NFA total under its account convention	gha	aggregate
E_{prod}	NFA Global Ecological Footprint of Production	gha	empirical account
E	model-defined footprint demand, eP (or exogenous fixed demand)	gha	aggregate
$X,\ C^{\mathrm{emp}}$	crop-proxy account $X = \hat{b}_f A_f$ and residual $C^{\mathrm{emp}} = B - X$; neither is physical capital	gha	illustrative accounting split
C^{mod}	model observation residual $B - b_f A_f = \Upsilon_c(q)A_c$, distinct from the empirical residual	gha	model-derived
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\mathrm{conv}}$	$\mathrm{ha\,yr^{-1}}$	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	$\mathrm{gha\,cap^{-1}}$	parameter
r	intrinsic population growth rate	$\mathrm{yr^{-1}}$	parameter
ρ_c	quality-regeneration scenario rate	$\mathrm{yr^{-1}}$	scenario parameter; not estimated
$q_{\max}$	imposed quality admissibility bound	dimensionless	not an estimated threshold
μ	cropland maintenance rate (reserve→fast)	$\mathrm{yr^{-1}}$	parameter
η_f	fast-land retirement rate (fast→reserve)	$\mathrm{yr^{-1}}$	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt in the multiplicative law	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
α_{add}	coefficient in the additive comparator $b_f = b_{f0} + T_b - \alpha_{\mathrm{add}} D$	$\mathrm{ha^{-1}\,yr^{-1}}$	comparator only; distinct from α
η	debt-repayment rate	$\mathrm{yr^{-1}}$	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delay scenarios	yr	scenario parameters
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter

Symbol	Meaning	Unit	Role
R_{rc}	surplus-gated restoration, $\chi_r A_r [B-E]_+$ (reserve→capital)	$\mathrm{ha\,yr^{-1}}$	gated flow
χ_r	restoration coefficient	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1+e^{-\kappa_w(t-t_{wave})})$	$\mathrm{gha\,ha^{-1}}$	off except in SI S8
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
Δb, κ_w, t_{wave}	amplitude, steepness and center of the wave	$\mathrm{gha\,ha^{-1}}$; $\mathrm{yr^{-1}}$; yr	active only in SI S8
$A_c^{\min}$	capital-land area floor	ha	enforced by conversion-rate law
$A_r^{\min}$	reserve monitoring threshold (0.10)	ha	monitored post hoc, not enforced
$A_f^{\max}$	optional fast-land ceiling	ha	not active in reported runs
$K_{\min}$	carrying-capacity floor	cap	imposed in definition of K
A_{tot}	total bioproductive area	ha	parameter
k	$\mu/(\mu+\eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived
$B_{eq}(A_c)$	maintenance-balanced land relation; equality at $B_{eq}=E$, with surplus no-conversion states when $B_{eq}>E$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	$\mathrm{yr^{-1}}$	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-land shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T)-A_c^{\mathrm{deg}})/(A_c^{\mathrm{init}}-A_c^{\mathrm{deg}})$	dimensionless	derived
R_{fc}, χ	fast→capital restoration variant, $\chi A_f [B-E]_+$	$\mathrm{ha\,yr^{-1}}$; $\mathrm{gha^{-1}\,yr^{-1}}$	extension (off)
L_c, H_c	timber liquidation flow $[H_c-\sigma_c Y_c]_+ / b_{G,c}$ and its target	$\mathrm{ha\,yr^{-1}}$; gha	extension (off)

Model extensions. The reported simulations do not include timber liquidation or direct restoration from fast land to capital land. SI §S6 considers reservations of the human-available shares as a policy extension. The technology wave is off in the baseline and gate-sign comparison; it is switched on only for the matched experiment in SI §S8.

Conventions. The positive-part operator is exact, conversion runs at the rate set by τ_{conv}, and transfers conserve area across A_f, A_c, and A_r. The quality index q is an uncalibrated regeneration/readout state. Larger q does not imply larger $\Upsilon_c(q)$: the latter values a growth flow, peaks at $q_{\max}/2$, and returns to b_c at $q_{\max}$. This observation rule is not a monotone physical-condition index. Its continuous specification is $K_q = [0, q_{\max}]$ and $\dot{q}(t) = \Pi_{T_{K_q}(q(t))} G_c(q(t-\tau_g))$,

where $G_c(x) = \rho_c x(1 - x/q_{\max})$, $\rho_c > 0$, and $q_{\max} > 0$. We use an admissible continuous history $q(s) = \phi(s) \in K_q$ for $s \in [-\tau_g, 0]$ and interpret the equation for an absolutely continuous solution almost everywhere. The baseline $q_{\max} = 1$ is imposed as an admissibility bound, not an estimated ecological threshold; ρ_c and τ_g are scenario settings, not estimates from recovery studies. Since $G_c(x) \geq 0$ on K_q, the quality state is nondecreasing; outward velocity is projected to zero at the upper bound and the lower bound is invariant. Maintenance at rate μA_r allows fast-land area to remain positive at equilibrium. The numerical update enforces $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in K_q$, $P, D \geq 0$, and $K \geq K_{\min}$; reserve area is reconstructed from total area, without projection onto $A_r \geq A_r^{\min}$. The reserve threshold is monitored post hoc, not an invariant of the displayed flow. In the reported $\Delta t = 0.1\,\mathrm{yr}$ drawdown grid, the minimum reserve was 0.511 ha (initial reserve is 0.72 ha for $A_{c0} = 1.18\,\mathrm{ha}$); the gate-comparison minimum was 0.602 ha. These tested values exceed $A_r^{\min} = 0.10\,\mathrm{ha}$ but do not prove invariance. The optional $A_f^{\max}$ ceiling is not imposed in reported runs.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiate $B = b_f A_f + \Upsilon_c(q) A_c$, use $b_f = (b_{f0} + T_b)e^{-\alpha D}$, and substitute Eqs. (6)–(7) to obtain the general identity

$$\dot{B} = \Delta b_{\mathrm{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c} G_c'(q) A_c\, \dot{q} + \Upsilon_c R_{rc} - \alpha \dot{D}\, b_f A_f + \frac{\dot{T}_b}{b_{f0} + T_b}\, b_f A_f. \quad \text{(S1)}$$

The conversion contribution has the sign of Δb_{conv}. The quality term follows the sign of $G_c'(q)$: it is non-negative for $q \leq q_{\max}/2$ and non-positive above that point, since $\dot{q} \geq 0$. While debt accumulates, its contribution is non-positive; a rising technology wave and restoration contribute non-negatively. Proposition 1 in the main text sets $T_b \equiv 0$, $R_{rc} \equiv 0$, and holds D fixed. In the demonstration run, D instead rises to $0.20\,gha\,yr$ and then declines. Initially, conversion contributes $0.21\,\mathrm{gha\,yr^{-1}}$ to $\dot{B}$, compared with $-0.005\,\mathrm{gha\,yr^{-1}}$ from fast-land maintenance and retirement and a debt contribution of about $-0.004\,\mathrm{gha\,yr^{-1}}$. Conversion therefore dominates the initial change in aggregate model biocapacity.

S2.2 The deficit-side affine system and its spectrum

On the deficit side, set $q = q_{\max}$, freeze D, switch off R_{rc}, and assume $S/b_f < A_c - A_c^{\min}$. Equation (5) then gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$. Using $A_r = A_{tot} - A_f - A_c$, Eqs. (6)–(7) reduce to

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \tag{S2}$$

The matrix has trace $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and determinant $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$, where $k = \mu/(\mu + \eta_f)$. At baseline, $\operatorname{tr} J = -1.0512\,\text{yr}^{-1}$ and $\det J = 0.05353\,\text{yr}^{-2}$. Its eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\text{yr}^{-1},\ -0.9975\,\text{yr}^{-1}. \tag{S3}$$

Because J depends on neither demand E nor the current state, this spectrum stays constant along the fixed-demand deficit-side branch. At $E/E_{\text{ceil}} = 0.50, 0.90, 0.99, 0.999$, the slow eigenvalue remains $\lambda_1 = -0.05366\,\text{yr}^{-1}$. It is unchanged as ρ_c varies ($\lambda_1 = -0.0537\,\text{yr}^{-1}$ at $\rho_c = 0.04, 0.08, 0.16\,\text{yr}^{-1}$), shifts modestly with τ_{conv} (-0.0553, -0.0537, -0.0505, $-0.0417\,\text{yr}^{-1}$ at $\tau_{\text{conv}} = 0.5, 1, 2, 5\,\text{yr}$), and shifts more with μ (-0.0254, -0.0537, $-0.1099\,\text{yr}^{-1}$ at $\mu = 0.03, 0.06, 0.12\,\text{yr}^{-1}$). It remains negative in every tested sweep.

The affine deficit-side system has the fixed point $A_c^*(E)$ from Eq. (18), with $A_f^* = k(A_{tot} - A_c^*)$. For $E > E_{\text{ceil}}$, this point lies below $A_c^{\min}$; the trajectory instead approaches $A_c = A_c^{\min}$ asymptotically, with $u_c \to 0$ and persistent shortfall $S^* = E - E_{\text{ceil}}$. Separately, in the full endogenous-population model with $K_{\min}$ inactive, a positive stationary population requires $E = B$, so its no-conversion equilibria lie on the equality relation with $P = B/e$. Two such baseline examples are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\text{ha}, 0.824\,\text{ha}, 0.762\,\text{ha}, 1.488\,\text{cap})$ and $(1.091\,\text{ha}, 0.500\,\text{ha}, 0.909\,\text{ha}, 1.731\,\text{cap})$. Both satisfy $A_r^*/A_f^* = \eta_f/\mu = 0.8333$; the second is a sustainable cropland equilibrium supported by maintenance. These full-system equality equilibria are distinct from the fixed-demand reduced surplus family ($B_{eq} > E$) and from the one-sided deficit-branch spectrum. The reserve-share mode explains why the strict-inequality interval in the delayed-quality demonstration lasts about 149 yr, even though most conversion is complete within 25 yr.

When the other branch of the minimum applies ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$. The distance to the floor then follows $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it shrinks exponentially but never reaches zero in finite time. The floor is approached asymptotically, so operational proximity thresholds depend on the tolerance used to define "near" (Section S4.2).

S2.3 Quality-bound and population-delay scope

The projected quality law is a constrained delay equation, so the characteristic roots of the raw, unconstrained DDE do not describe its boundary dynamics. For any admissible history and $\rho_c > 0$, $G_c(q(t - \tau_g)) \geq 0$ on K_q, and the projected trajectory is nondecreasing. If $q(0) > 0$, it has a limit $L \geq q(0) > 0$. If $L < q_{\max}$, then $G_c(q(t - \tau_g)) \to G_c(L) > 0$, contradicting convergence to a finite interior limit; hence $q(t) \to q_{\max}$. The all-zero history remains at zero, so $q_{\max}$ is not a global attractor over every admissible history. We make no characteristic-root stability claim for the projected quality component at the active upper bound.

The population equation delays carrying capacity, not population in its own crowding term. If K is constant, the equation reduces to the ordinary logistic ODE; at a positive equilibrium $P^* = K^*$, its scalar population perturbation satisfies $\delta\dot{P} = -r\,\delta P$ and has eigenvalue $-r$. With a varying capacity, the local equation at $P^* = K^*$ is $\delta\dot{P} = -r\,\delta P + r\,\delta K(t - \tau_p)$: the delay is in the capacity input. Thus $r\tau_p < \pi/2$ is not a stability criterion for the equation written in the

manuscript. We do not report a full coupled characteristic-root classification for the projected two-land system.

For completeness, on a deficit-active face with $z = (A_f, A_c, q, P, D)$ and $b_f = (b_{f0} + T_b)e^{-\alpha D}$, the debt-state Jacobian row is

$$\frac{\partial \dot{D}}{\partial z} = \left(-b_f, -\Upsilon_c, -b_{G,c} G_c'(q) A_c, e, \alpha b_f A_f - \eta\right).$$

In particular, $\partial \dot{D}/\partial D = \alpha b_f A_f - \eta$; the feedback through $b_f(D)$ is part of the derivative. This row describes the smooth interior of the deficit-active face and is not a stability classification at the quality ceiling or the shortfall switch.

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\text{ceil}}$, capital-land area approaches $A_c = A_c^{\min}$ and the persistent shortfall feeds debt. At the floor, $B_{eq}(A_c^{\min}; D) = b_{f0} e^{-\alpha D} k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, so $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$. Its fixed point $D^*(E)$ satisfies $\eta D = E - B_{eq}(A_c^{\min}; D)$. The right-hand side is increasing in D, making the fixed point unique and attracting when $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$; at baseline, $0.034\,\text{yr}^{-1} < 0.05\,\text{yr}^{-1}$. Using the unrounded demands $E = E_{\text{ceil}} + 0.02 = 1.158409091$ and $E = E_{\text{ceil}} + 0.10 = 1.238409091$ gha gives $D^* = 1.209655$ and 5.394538 gha yr. The floor-state integration uses $A_c(0) = A_c^{\min}$, $A_f(0) = k(A_{tot} - A_c^{\min})$, $A_r(0) = (1-k)(A_{tot} - A_c^{\min})$, $q(0) = q_{\max}$, $D(0) = 0$, $\Delta t = 0.1$ yr, and $T = 1200$ yr; its final debt agrees to the shown precision. The corresponding $B(D^*)$ values are 1.097926 and 0.968682 gha, with persistent shortfalls $S^* = 0.060483$ and 0.269727 gha. Under fixed demand, debt therefore saturates; it is unbounded only if the shortfall is unbounded.

S2.5 Boundedness asymmetry and the crossover shortfall

Equation (12) gives $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The technology term is bounded between $\ln b_{f0}$ and $\ln(b_{f0} + \Delta b)$, so its maximum contribution to $\ln b_f$ is $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ at baseline. From Eq. (13), $\dot{D} \le \bar{S} - \eta D$, where $\bar{S}$ is the running supremum of shortfall. Thus $D(t) \le \max\{D(0), \bar{S}/\eta\}$ and the log-debt term is bounded below by $-\alpha \max\{D(0), \bar{S}/\eta\}$. If $D(0) = 0$, this reduces to $-\alpha \bar{S}/\eta$. For a sustained shortfall with limit S^*, $D \to S^*/\eta$; its asymptotic debt loss equals the maximum technology log gain at $S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296$ gha. This is a sustained-shortfall comparison, not an exact transient equality for every run.

The fast-land contribution $Y_f = b_f A_f$ has the two direct yield channels described above. At fixed A_c and q, the capital contribution $\Upsilon_c A_c$ has no direct dependence on the technology increment or debt-related fast-land yield factor; indirect effects through the coupled land and demand dynamics remain possible. Under the additive comparator $b_f = b_{f0} + T_b - \alpha_{\text{add}} D$, the effects are directly comparable as bounded yield increments (Δb versus $\alpha_{\text{add}} D^*$), so the logarithmic asymmetry disappears. Here $[\alpha_{\text{add}}] = \text{ha}^{-1}\,\text{yr}^{-1}$, distinct from the multiplicative α in Eq. (12). The bound above applies to the multiplicative specification.

S2.6 Macro-ratio identities

Set $R_B = E/B$, $R_A = E/Y_f$, and $\psi_f = Y_f/B$. Then $R_A = R_B/\psi_f$ by definition. The direct-yield share of capital contribution, $\psi_c = b_c A_c/Y_c$, is a different ratio; $R_A = R_B/\psi_c$ is not an identity. Aggregate balance occurs at $R_B = 1$. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ is necessary but not sufficient for conversion: shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$. The threshold changes when flows are reserved. Capital-land area falls when $u_c > R_{rc}$, with u_c defined in Eq. (5). Thus, under baseline shares, conversion requires $R_B > 1$. Under the reservation extension with $\sigma_f, \sigma_c < 1$, conversion and declining A_c can occur even when $R_B \le 1$; aggregate overshoot is not required. At $q^* = q_{\max}/2$, the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at baseline. This value applies at that point; it is not a structural constant.

S3 Numerical methods and sensitivity analyses

Numerical method and finite-step implementation check. The reported two-land simulations use the projected delayed-quality law in Eq. (10), integrated by the method of steps with explicit first-order Euler updates. At step k, current states are read at t_{k-1}; the time-dependent technology input and delayed values are evaluated at t_k, with linear interpolation of the delayed q and B histories. The raw delayed quality velocity is projected onto $T_{K_q}(q(t_{k-1}))$ before the Euler update. After the state update, q is clipped to $K_q = [0, q_{\max}]$ as a finite-step safeguard; the explicitly enforced state bounds are applied, reserve area is reconstructed as $A_r = A_{tot} - A_f - A_c$, and the updated B is stored in the delayed history. The reserve threshold is monitored, not projected. The positive-part operator is exact. For ordinary runs, $q(s) = q_0$ and $B(s) = B_0$ over the full pre-run interval $[-\max(\tau_g, \tau_p), 0]$, with B_0 calculated from the initial state. In the technology-wave experiment, both runs share a no-wave prehistory; the active $T_b(0)$ changes current $B(0)$, and later wave-adjusted B values enter $K(t - \tau_p)$. Baseline demonstrations and threshold bisections use $\Delta t = 0.05\,\mathrm{yr}$; drawdown sweeps use $0.1\,\mathrm{yr}$, with selected thresholds rechecked at $0.05\,\mathrm{yr}$.

A finite-step consistency test compared post-Euler quality clipping with explicit tangent-cone projection before the Euler update. For the Figure 1 initial state and parameters, $T = 100\,\mathrm{yr}$, $\Delta t = 0.05\,\mathrm{yr}$, and $\tau_g = 5, 20, 40\,\mathrm{yr}$, the arrays for A_f, A_c, q, P, D, A_r, B, E, and T_b were exactly equal. For the continuous baseline projected-quality path at $\tau_g = 20$ or $40\,\mathrm{yr}$, $q(t) = \min(1, 0.9 + 0.0072t)$; its maximum piecewise-linear error fell from 7.1×10^{-5} at $\Delta t = 0.1\,\mathrm{yr}$ to 8.9×10^{-6} at $0.0125\,\mathrm{yr}$. In the full fixed-demand Figure 1 scenario ($P_0 = 1.5\,\mathrm{cap}$), halving the step from 0.05 to $0.025\,\mathrm{yr}$ reduced maximum absolute differences against a $0.003125\,\mathrm{yr}$ reference from 0.00221 to $0.00102\,\mathrm{ha}$ for A_c, 0.00180 to $0.000831\,\mathrm{gha}$ for B, 0.000245 to $0.000114\,\mathrm{cap}$ for P, and 0.00200 to $0.000924\,\mathrm{gha\,yr}$ for D. These checks establish equality only for the enumerated cases and show smaller differences under timestep refinement; they are not a universal error bound or proof of convergence for all settings.

Selected delayed law and current-quality sensitivity. The selected model uses $\dot{q} = \Pi_{T_{K_q}(q(t))} G_c(q(t - \tau_g))$. A current-quality comparator, with the same interval constraint but $G_c(q(t))$, is included only as a sensitivity check and is not the selected model. Under the baseline constant history $q(s) = 0.9$, $\rho_c = 0.08\,\mathrm{yr}^{-1}$, and $q_{\max} = 1$, the projected delayed-quality paths for $\tau_g = 20$ and $40\,\mathrm{yr}$ are exactly identical: both reach the imposed ceiling at $125/9 \approx 13.8889\,\mathrm{yr}$, before either delay elapses. This is saturation-induced degeneracy, not general delay robustness; the $\tau_g = 5\,\mathrm{yr}$ case is distinct. The current-quality comparator extends the demonstration window by $1.3\,\mathrm{yr}$ and shifts the floor-proximity threshold by at most $0.001\,\mathrm{cap}$ in the displayed finite comparison. This finite comparison does not imply equivalence between the two quality laws.

Well-posedness and scope. For $\rho_c > 0$, $q_{\max} > 0$, and an admissible continuous history, the delayed input $G_c(q(t - \tau_g))$ is nonnegative. On each method-of-steps interval for $\tau_g > 0$, this input is known from the preceding interval, and the projected solution is the unique absolutely continuous path obtained by integrating that input until $q_{\max}$ is reached and then holding $q = q_{\max}$. Iterating gives an invariant quality path in K_q; for $\tau_g = 0$ the scalar projected ODE has the same unique constrained solution. If $q(0) > 0$, it converges to $q_{\max}$; the all-zero history remains zero. The boundary projection makes the vector field constrained and nonsmooth at $q = q_{\max}$, so we do not transfer characteristic roots of the raw unconstrained delay equation to the projected system. The other right-hand sides are locally Lipschitz on each specified active set, and the positive-part and minimum operators are Lipschitz. Area conservation is maintained by reconstructing reserve area; the reserve monitoring threshold is not projected, as specified in SI §S1. The affine spectrum reported in SI §S2 applies only to the one-sided deficit-active land branch; no smooth linear-stability claim is made across the quality ceiling or the shortfall switch.

Table S2: Current-quality sensitivity comparator versus the selected projected delayed-quality specification. The comparator uses the same imposed interval K_q; reported baseline trajectories use the delayed law. The equality of the delayed-quality $\tau_g = 20$ and 40 yr paths is saturation-induced under the stated history, not evidence of general delay robustness. B, Y_f, and $\Upsilon(q)A_c$ are in gha; land areas in ha; P_0 in cap; delays and step intervals in yr.

Quantity	current $q(t)$	projected delayed $q(t-\tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial → mask end, gha)	0.08364 → 0.04038	0.08364 → 0.04038
Fast-land contribution Y_f (initial → mask end, gha)	0.51000 → 0.78462	0.51000 → 0.78462
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 5$ yr, $\Delta t = 0.1$ yr)	2.955	2.955
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 20$ yr, $\Delta t = 0.1$ yr)	2.955	2.954
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 40$ yr, $\Delta t = 0.1$ yr)	2.955	2.954

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 reports the full fast-land-yield sweep summarized in the main-text fixed-demand results, including the balance-branch slope. The sign of Δb_{conv} stays positive when $b_{G,c}$ triples. At $b_f = 0.85\,\text{gha}\,\text{ha}^{-1}$, $\Upsilon_c(q^*) = 0.058, 0.066, 0.074\,\text{gha}\,\text{ha}^{-1}$ for $b_{G,c} = 0.4, 0.8, 1.2\,\text{gha}\,\text{yr}\,\text{ha}^{-1}$, giving $\Delta b_{\text{conv}} = +0.792, +0.784, +0.776\,\text{gha}\,\text{ha}^{-1}$.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\text{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 reports the minimum capital-land area reached in endogenous-demand runs over a grid of initial populations P_0. The runs use $A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$, $\Delta t = 0.1$ yr, and $T = 400$ yr; the full P_0 grid (step 0.1) is in `data/si_drawdown_depth_C18_A01.csv`. Depending on P_0, the minimum occurs 3–25 yr after the run begins, when the shortfall first closes ($B \geq E$). At $A_{c0} = 1.00$ ha, the delayed-quality 10^{-6} ha proximity values shift by at most 0.001 cap across the tested $\tau_g = 5, 20, 40$ yr cases (Table S2). For the baseline history, the $\tau_g = 20$ and 40 yr quality paths are exactly identical because both reach the imposed cap before either delay elapses; this is saturation-induced degeneracy, not general delay robustness. The $\tau_g = 5$ yr case is distinct. Across the tested finite grid, drawdown depth varies monotonically with both P_0 and A_{c0}. The grid and selected bisections show no discontinuity at the tested resolution but do not establish a general continuity theorem.

The initial populations at which the minimum comes within 10^{-2}, 10^{-3}, and 10^{-4} ha of the floor are 1.8, 1.9, 2.1 cap for $A_{c0} = 0.40$ ha; 2.2, 2.3, 2.4 for 0.70 ha; 2.6, 2.7, 2.8 for 1.00 ha; and 2.8, 2.9, 3.0 for 1.18 ha. Bisection with a 10^{-6} ha proximity tolerance ($\Delta t = 0.05$ yr, $T = 1000$ yr) gives 2.41, 2.70, 2.98, 3.14 cap. The thresholds match to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.954 cap at $A_{c0} = 0.40$ and 1.00 ha, $\Delta t = 0.1$ yr).

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05\,\text{ha}$.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

Values printed as 0.050 ha in Tables S4–S5 are rounded values above the floor, not finite-time floor hits. The full-precision CSV reports $\min A_c - A_c^{\min}$ explicitly. For example, at $A_{c0} = 1\,\text{ha}$ and $P_0 = 3$, the minimum exceeds the floor by $4.60 \times 10^{-7}\,\text{ha}$; finite arithmetic can eventually remove still smaller gaps. The asymptotic statement concerns the continuous conversion law, not exact preservation of arbitrarily small positive gaps by floating-point arithmetic.

S4.3 Recovery metric and the role of the restoration flow

We measure recovery as $\mathcal{R}_c(T) = (A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$. A value of 0 means no recovery, 1 means return to the reference, a negative value means further decline, and a value above 1 means over-restoration. Table S5 applies this metric to delayed-quality runs with $A_{c0} = 1.0\,\text{ha}$ ($A_c^{\text{init}} = 1.0\,\text{ha}$, $A_c^{\text{deg}} = \min_t A_c$, $T = 400\,\text{yr}$). Without restoration, A_c stays at its minimum because no return flow exists. With surplus-gated restoration, recovery is partial in all six initial-population cases ($\mathcal{R}_c = 0.21$–0.89); none returns to the reference. The separate gate-sign experiment in SI §S4 uses other initial states and interventions, including cases of over-restoration.

Table S5: Recovery of the capital-land area after a demand-driven drawdown ($A_{c0} = 1.0\,\text{ha}$, $A_{f0} = 0.6\,\text{ha}$, $q_0 = 0.9$, $T = 400\,\text{yr}$): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless. The timestep is $\Delta t = 0.1\,\text{yr}$; S10 gives timestep-refinement results for $P_0 = 2$.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	0.969	0.89
2.0	1.100	0.398	3.8	0.398	0.677	0.46
2.5	1.375	0.075	4.1	0.075	0.433	0.39
3.0	1.650	0.050	13.8	0.050	0.328	0.29
3.5	1.925	0.050	20.6	0.050	0.278	0.24
4.0	2.200	0.050	26.6	0.050	0.250	0.21

S4.4 Gate-sign phase dependence

The gate-sign simulations start from a non-equilibrium post-drawdown state: $A_c = 0.058\,\text{ha}$, $A_f = 0.6\,\text{ha}$, $q = 0.9$, $P = 1.2\,\text{cap}$, and $D_0 = 0\,\text{gha yr}$. Area conservation gives $A_r = 1.842\,\text{ha}$. With the baseline parameters and $T_b = 0$, the initial account amount is $B_0 = 0.513234\,\text{gha}$, while $E_0 = eP_0 = 0.660\,\text{gha}$; hence the run begins in shortfall ($S_0 = 0.146766\,\text{gha}$ and $R_B = 1.286$), not at an equilibrium. The deficit-gated restoration is therefore initially active, while surplus-gated restoration is initially inactive. This starting point is distinct from the separate degraded equilibrium, where $R_B = 1$ and $B = E$, so both restoration gates are zero. The healthy reference equilibrium has $A_c^{\text{init}} = 0.8225\,\text{ha}$ and $P^{\text{init}} = 1.489\,\text{cap}$; the degraded reference used in $\mathcal{R}_c$ is $A_c^{\text{deg}} = 0.0513\,\text{ha}$. Every row uses this same stated initial state and $T_b = 0$.

Table S6: Restoration gate outcomes. $\mathcal{R}_c$ is reported at $T = 100$ and 900 yr; final-state columns A_c, A_f, A_r, B, and P/P^{init} are at $T = 2000$ yr. The extrema $\min A_r$, $\max A_c$, and peak debt are computed over the inclusive interval $0 \le t \le 2000$ yr. Annual account values $B = E$ are in gha; land areas are in ha; population ratios and recovery metrics are dimensionless. Each row approaches $R_B = 1$ ($E = B$).

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.058	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.097	0.765	0.638	0.705	0.86	0.602	1.097	0.543
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

Without restoration, A_c stays at its minimum because the model has no return flow. Surplus-gated restoration produces partial or over-restoration depending on χ_r. At $\chi_r \ge 0.05$, it continues while $B > E$, and supported population falls to 0.97–0.90 of its reference value. A target cap returns A_c to its reference ($\mathcal{R}_c = 1.00$). Deficit-gated restoration acts during shortfall, but at the tested rates it is weaker than concurrent conversion, leaving recovery near zero. Freezing conversion under surplus-gated restoration raises peak debt from 0.152 to 0.543 gha yr; restoration then shifts reserve land to capital use while fast-land maintenance declines. These outcomes depend on the starting state and intervention rates. The gate-sign comparison keeps $T_b = 0$; SI §S8 reports the separate matched wave/no-wave debt experiment.

Policy implementation and numerical refinement. The gate experiment uses the selected projected delayed-quality law, $\Delta t = 0.1$ yr, and constant initial-state histories over both delays. Reference states are obtained from no-restoration runs to 3000 yr with $(A_{c0}, P_0) = (0.9, 1.2)$ and $(0.9, 2.5)$; their reference values are held fixed when the policy integration step is refined. The target rule limits restoration by the remaining target gap per step. Conversion-freeze rows set $u_c = 0$ independently of the gate. Full state extrema include $t = 0$. The versioned output reports all nine policies at $\Delta t = 0.1, 0.05, 0.025$ yr, including the separate final B and E rather than enforcing their equality. Refinement quantifies numerical sensitivity; it is not field validation.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutrality	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at the conversion margin; conversion alone cannot produce a composition mask

S5 Empirical details

S5.1 Data and vintages

Our empirical source is the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The workbook spans 1961–2025, but we use complete world observations through 2023 and exclude the estimated 2024–2025 rows. The 1961–2022 sensitivity uses the same edition and population series. This release recalculates historical values and reports five biocapacity account categories: fishing grounds, built-up land, cropland, grazing land, and forest biocapacity. These categories are account entries, not a common-boundary inventory of physical stocks.

The workbook's `pop_1000_persons` field reports UN PopStat population in thousands; we convert it to persons before calculating per-capita values. The field `efp_total_gha` gives the Global Ecological Footprint of Production, denoted E_{prod}. We calculate per-capita values from unrounded account totals and population rather than the workbook's rounded columns. Table S8 summarizes the endpoints. From 1961 to 2023, B rises from 9,725,417,562 to 12,142,843,961 gha, population from 3.06487 to 8.09173 billion, and B/P falls from 3.17319 to 1.50065 gha person^{-1}. Over the same period, E_{prod}/P rises from 2.19700 to 2.56462 gha person^{-1}. The identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ holds to rounding; it separates extensive and intensive changes but does not attribute causes. In both windows, 1973 is the first year with $E_{\text{prod}}/B > 1$.

Table S8: World endpoint changes in NFA Edition 2026. The 1961–2022 column is a same-edition sensitivity; 2024–2025 estimates are excluded. Population and per-capita quantities are recomputed from workbook totals.

Measure	1961–2023	1961–2022
$\Delta B/B_{1961}$	+24.8568%	+24.5450%
$\Delta \ln B$	+0.221997	+0.219497
$\Delta P/P_{1961}$	+164.0154%	+161.7211%
$\Delta \ln P$	+0.970837	+0.962109
$\Delta(B/P)/(B/P)_{1961}$	−52.7085%	−52.4131%
$\Delta \ln(B/P)$	−0.748840	−0.742612
$\Delta(E_{\text{prod}}/P)/(E_{\text{prod}}/P)_{1961}$	+16.7327%	+17.6994%
First year $E_{\text{prod}}/B > 1$	1973	1973

Annual transformation and precision. The offline pipeline selects exactly one world observation for each year 1961–2023 and joins the two external world series by year, with missing or duplicate records rejected. The annual extract includes population in persons, account components, indices, log changes, X, and the empirical residual. Source values are not rebalanced: the largest absolute difference between the sum of the five supplied components and the supplied total is 2 gha across the annual series (relative discrepancy below 2×10^{-10}), consistent with integer rounding in the source workbook. Individual percentage-point contributions are rounded independently; consequently the printed 2022 components sum to 24.5451 rather than the independently rounded 24.5450 total. Figures 2–3 in the main article use this same annual pipeline. NFA inputs and NFA-derived numerical outputs retain the source's CC BY-SA 4.0 attribution; external physical area and cereal yield retain OWID/FAOSTAT attribution and provider terms.

S5.2 Divisia decomposition, sensitivity to the base-year share and level fallback

The crop proxy is an illustrative account construction. We combine independently sourced OWID/FAOSTAT world cropland area, defined as arable plus permanent-crop area (ha), with cereal yield (tonnes per hectare) (Our World in Data, 2026b,a). Cereal yield is a productivity indicator, not a measure of output from all crops. For a declared 1961 account-share anchor ϕ_0, we set $\hat{b}_f(t) = \hat{b}_f(1961)\tilde{b}_f(t)/\tilde{b}_f(1961)$, with $\hat{b}_f(1961)A_f(1961) = \phi_0 B(1961)$. We then define $X_t = \hat{b}_f(t)A_f(t)$ and $C_t^{\text{emp}} = B_t - X_t$. Thus X is an index-scaled account proxy, and C^{emp} is the residual that reconciles it to the NFA total. The residual is not observed ecological capital, non-cropland biocapacity, or the model quantity $\Upsilon_c(q)A_c$.

When both X and C^{emp} are positive at the two endpoints, the exact log-mean Divisia identity is

$$\Delta \ln B = w_X\left(\Delta \ln \tilde{b}_f + \Delta \ln A_f\right) + w_C \Delta \ln C^{\text{emp}}, \qquad w_X = \frac{L(X_T, X_0)}{L(B_T, B_0)}, \quad w_C = \frac{L(C_T^{\text{emp}}, C_0^{\text{emp}})}{L(B_T, B_0)}, \tag{S4}$$

Here $L(a, b) = (b - a)/(\ln b - \ln a)$ is the logarithmic mean, with its continuous limit when $a = b$. We use the measured 1961 NFA cropland share, $\phi_0 = 0.1286472202$, as the central anchor. Mapping that account share to the external area–yield proxy is an assumption, not an independent observation of physical capital. Table S9 gives the selected sensitivity for 1961–2023. At the central anchor, the cereal-yield, area, and residual terms are +0.268662, +0.037710, and −0.084375, summing to $\Delta \ln B = +0.221997$. The same-edition 1961–2022 sensitivity gives +0.264696, +0.037506, and −0.082705, summing to +0.219497. The decomposition is exact for this proxy, but its allocation depends on ϕ_0 and is not a physical-stock attribution.

Table S9: Sensitivity of the proxy Divisia decomposition to the 1961 account-share anchor ϕ_0 (world, 1961–2023). Contributions and weights are dimensionless. The $\phi_0 = 0.128647$ row uses the measured NFA cropland share in 1961; other rows are illustrative calibrations.

ϕ_0	w_X	w_C	Cereal yield	Area	Residual C^{emp}	Sum
0.100000	0.1833	0.7957	+0.2088	+0.0293	−0.0162	+0.2220
0.128647	0.2358	0.7352	+0.2687	+0.0377	−0.0844	+0.2220
0.150000	0.2749	0.6893	+0.3133	+0.0440	−0.1352	+0.2220
0.190000	0.3483	0.6008	+0.3968	+0.0557	−0.2305	+0.2220
0.250000	0.4582	0.4581	+0.5221	+0.0733	−0.3734	+0.2220
0.300000	0.5499	0.3178	+0.6265	+0.0879	−0.4924	+0.2220
0.320000	0.5866	0.2454	+0.6683	+0.0938	−0.5401	+0.2220
0.340000	0.6232	0.1009	+0.7100	+0.0997	−0.5877	+0.2220

For $\phi_0 \geq 0.340531$ in 2023 (or 0.343400 in 2022), terminal residual C^{emp} is nonpositive, so logarithmic weighting is invalid. We use the exact level identity $\Delta B = \Delta X + \Delta C^{\text{emp}}$ instead;

this is still an accounting identity, not a stock measure. At the selected anchor, $X_{2023}/B_{2023} = 0.377784$ and C^{emp} remains positive, but the proxy fast-land share and residual contribution vary across the sensitivity analysis. Every row uses the same NFA total B. The residual absorbs differences between the index-scaled crop proxy and that total, including account categories, boundaries, valuation, vintage, and calibration.

S5.3 Cropland-account calibration

We report NFA cropland-account movements separately from the illustrative proxy. From 1961 to 2023, NFA cropland biocapacity rises 2.870055-fold and NFA cropland area rises 1.198029-fold; their account-value-per-area quotient rises 2.395647-fold. For 1961–2022, the same-edition ratios are 2.851441, 1.198326, and 2.379520. This quotient is an NFA account value per area, not an independent physical crop-yield measure. Over 1961–2023, the external OWID/FAOSTAT area series rises 1.173414-fold and cereal yield rises 3.124667-fold. Those independent series serve as inputs to an illustrative proxy; they do not reconstruct the NFA cropland account or measure output from all crops. The model's representative $b_{f0} = 0.85\,\text{gha}\,\text{ha}^{-1}$ remains a conditional parameter, not an empirical fit to these ratios.

S5.4 Observation-operator hierarchy

- *NFA accounting observations.* Total and component values in gha describe NFA account composition and change under the stated edition and conventions. They do not directly measure biomass, soil condition, wetland extent, fish biomass, or model state A_c.
- *Fast-provisioning area.* The external world cropland-area series covers arable land plus permanent crops for 1961–2023 (ha) (Our World in Data, 2026b). It is an independent proxy input, not the NFA cropland-area series.
- *Crop-productivity indicator.* OWID/FAOSTAT cereal yield is reported in tonnes per hectare for 1961–2024 (Our World in Data, 2026a). It covers cereals, not all crops, and is not a physical-stock measure. Base-anchored $\hat{b}_f$ is used only in the illustrative account proxy.
- *Ecological-capital components.* Separate physical observation operators could use forest extent and biomass, soil carbon and condition, wetland extent and condition, and fish biomass by species or stock. Such data would need harmonized boundaries, reference years, definitions, uncertainty, and explicit treatment of overlap. We assembled no common-boundary global physical-stock series for these components. NFA forest biocapacity and residual C^{emp} cannot substitute for those measurements.
- *Model mapping.* The model defines capital contribution as $Y_c = \Upsilon_c(q)A_c$. Neither A_c nor q is identified empirically by total B or an NFA component, and we do not assume proxy residual $C^{\text{emp}} = B - X$ equals Y_c. A model–data comparison would first need independent physical measures and an explicit mapping.

Table S10 summarizes the possible links while keeping NFA categories distinct from model states.

S5.5 Regeneration anchors

The model's capital-land regeneration timescale represents a lumped recruitment response, not a universal ecological rate. The cited forest studies use chronosequence biomass records; fisheries studies analyze stock-specific biomass trajectories, often derived from assessment models; soil studies use different land-use and carbon-response definitions (Poorter et al., 2016; Hutchings

Table S10: Conceptual mapping from two-land model quantities to NFA information and independent proxy inputs. The rows specify possible observation operators, not identities between NFA categories and model states. Annual account values are in gha; land areas are in ha.

Model quantity	Closest account or proxy information	Interpretation and identification limit
Fast area and proxy account A_f, $X = \hat{b}_f A_f$	NFA cropland account; independently sourced OWID cropland area and cereal yield	NFA cropland biocapacity is an account value; the external area–yield product is an illustrative, base-anchored proxy. Neither is an observed model fast-land yield or a one-to-one physical measure.
Capital area, quality and contribution A_c, q, $Y_c = \Upsilon_c(q)A_c$	No direct NFA counterpart; the proxy residual $C^{\text{emp}} = B - X$ is defined by subtraction	In the model $Y_c = \Upsilon_c(q)A_c$; empirically C^{emp} is not assumed equal to Y_c. It also absorbs differences in boundaries, account categories (including built-up land), valuation, vintage and proxy calibration, and is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S11: Observability ladder: what is directly measured, what is identifiable by construction, and what remains unidentified.

Tier	Quantities	Status
Observable physical area input	OWID cropland area [ha]; other mapped land-cover classes where available	Measured extent, but mapping to effective A_c is not direct
Observable population	P [persons]	Direct in the UN PopStat workbook field after unit conversion
Observable productivity indicator	Cereal yield [tonnes ha^{-1}]	Physical yield of cereals only; not all-crops output or ecological capital
Observable annual account value	NFA B, E_{prod} and components [gha]	Measured only under the NFA accounting convention; value-weighted accounts
Proxy account and residual	$X = \hat{b}_f A_f$ and $C^{\text{emp}} = B - X$ [gha]	Identified by construction for the stated inputs and anchor; not physical stocks and not assumed equal to $\Upsilon_c(q)A_c$
Effective capital contribution	$Y_c = \Upsilon_c(q)A_c$ [gha]	A model quantity; no common-boundary physical observation operator is assembled here
Not separately identifiable	A_c, q, b_c, $b_{G,c}$, ρ_c and D	Requires independent physical-state, quality, regeneration or degradation observations and a declared mapping

and Reynolds, 2004; Neubauer et al., 2013; Poeplau et al., 2011). These sources document heterogeneous recovery behavior, but we do not refit a common recovery curve or compute harmonized t_{50} or τ_r values from their underlying data here.

These records describe different systems and recovery definitions; they are not pooled or used to assign a generic response time to the lumped capital-land class. The 10–40 yr interval is a scenario range, not an empirical estimate. Neither ρ_c nor τ_g is calibrated from these records, and they do not validate the present quality equation or parameter values.

S5.6 Recursive identifiability in the land-type accounts

NFA Edition 2026 illustrates why the aggregate should be read with its components, without treating account movements as physical-stock changes. From 1961 to 2023, cropland contributes +24.0577 percentage points to the +24.8568% change in total B; built-up land adds +7.4416, while Forest biocapacity contributes −5.9525, grazing land −0.2688, and fishing grounds −0.4212. The non-cropland sum, which includes built-up land, changes by only +0.7990 points (1.0092×) because unlike account movements offset one another.

For the same-edition 1961–2022 sensitivity, contributions are +23.8183 points for cropland, +7.2854 for built-up land, −5.8137 for Forest biocapacity, −0.2890 for grazing land, and −0.4559 for fishing grounds. The non-cropland sum changes by +0.7267 points (1.0083×). These are accounting attributions, not evidence that forest biomass, soil condition, wetland extent, fish biomass, or model A_c declined. Built-up land is an NFA account class, not ecological capital; including it in the non-cropland sum is arithmetic, not an ecological aggregation.

S5.7 Physical-stock measurement specification

We assembled no common-boundary global physical-stock series for these components, so we present no physical-stock composite or global decline finding. This does not mean that such a series could never be assembled. Table S12 is an unpopulated specification of candidate measures and comparability requirements; it contains no empirical endpoints and is not a model calibration.

These physical measures are heterogeneous. Forest biomass and soil carbon are masses; wetland and cropland extent are areas; fish biomass is stock-specific; freshwater storage is a volume; and condition indicators are often dimensionless or categorical. They are not dimensionally comparable and may overlap spatially. Annual flows—harvest, catch, increment, recharge, or emissions—must also remain distinct from stocks and extents. Even after normalization, interpretation depends on overlap, baselines, uncertainty, ecological thresholds, valuation, and allowed substitution. Unless common spatial and temporal boundaries, a normalization or valuation rule, and substitution rules are specified, the components should remain a nonadditive vector. A scalar composite is not neutral and cannot be inferred from NFA gha accounts. SEEA Ecosystem Accounting likewise distinguishes ecosystem extent and condition from ecosystem-service and monetary presentations (United Nations et al., 2021).

In the two-land model, A_c is an effective area state (ha) and q is a separate quality index; neither is observed in the NFA total. Physical-stock data could inform a model observation operator only after boundaries and mappings to these effective states are specified. Proxy residual $C^{\text{emp}} = B - X = B - \hat{b}_f A_f$ is not ecological capital, a measured non-cropland stock, or an estimate of $\Upsilon_c(q)A_c$. NFA components remain account values, not physical-stock observations.

Table S12: Physical-stock measurements and comparability requirements. The rows are unpopulated: no common-boundary global physical-stock series for the relevant components or endpoint values were assembled. Stocks, extents, condition measures, and annual flows are not combined into a composite.

Component	Physical stock, extent, or condition to define	Annual flow or pressure (separate)	Comparability rule required
Forests	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition	Net annual increment, harvest, or deforestation	Common forest definition and mapped area; reference year; treatment of plantations and overlap
Soils	Soil organic-carbon stock or density for a stated depth; structure or erosion condition	Carbon accumulation or loss; erosion where measured	Consistent depth, bulk density, land cover, and sampling method
Wetlands	Extent (ha) and an explicit ecological-condition measure	Carbon accumulation, water-storage change, or nutrient flux	Wetland definition and overlap with forest and other land-cover classes
Fisheries	Spawning-stock biomass by species or stock (tonnes); age or size structure	Recruitment, catch, and landings	Stock identity, assessment method, reference point, and uncertainty
Freshwater	Surface-water or groundwater storage (m^3), level, and quality	Recharge and withdrawal	Basin or aquifer boundary; separate stored volume from annual flow

S6 Policy extensions

Reserved biocapacity share. If both land classes have the same reservation share, $\sigma_f = \sigma_c = \sigma < 1$, human-available supply is σB and shortfall is $S = [E - \sigma B]_+$. Reservation lowers the demand at which conversion begins; conversion is off while $E \leq \sigma B$. With different shares, the condition becomes $E \leq \sigma_f Y_f + \sigma_c Y_c$, which need not equal σB. These results apply to the stated parameterization; they do not establish robustness to disturbances or parameter uncertainty.

Typed floors and conversion stop. The numerical model enforces $A_c \geq A_c^{\min}$ through its conversion-rate law and enforces $K \geq K_{\min}$ by definition. Reserve area is reconstructed from total area; $A_r^{\min} = 0.10\,\text{ha}$ is a post-hoc monitoring threshold, not an invariant projection. The optional $A_f^{\max}$ ceiling is not active in the reported runs. For $y = A_c - A_c^{\min} \geq 0$, Eq. (5) gives $0 \leq u_c \leq y/\tau_{\text{conv}}$; with $R_{rc} \geq 0$, $\dot{y} \geq -y/\tau_{\text{conv}}$, so $y(t) \geq y(0)e^{-t/\tau_{\text{conv}}}$. A stop at the exact floor is therefore an operational safeguard, not needed to establish this continuous-time bound. Total B cannot reconstruct A_c or implement arbitrary A_c-dependent policies, but that information limitation alone is not a universal control impossibility.

Institutional signature. Related work distinguishes protective and mobilizing institutional feedback (Abaee, 2026a). Here, surplus-gated restoration is off during shortfall, while deficit-gated restoration acts during shortfall and can further reduce provision (Section S4.4).

Aggregation and typed feasibility. A related transition-assessment study defines a linear compensatory operator and compares perfect substitutability with less-substitutable CES alternatives (Abaee, 2026b). ECOMOD's B is also a linear sum, but the companion considers a separate finite-plan setting. There, the compensatory protocol accepts states even when no available plan meets every component floor (compensatory-acceptance result in Abaee 2026b). The diagonal-gap analysis in Abaee (2026b) identifies an exact interval of diagonal-gap states on its witness-defined region that each positive elasticity falsely certifies as safe. Thus no positive-elasticity member is uniformly safe on that region. Under pathwise feasibility, falling below a typed floor during

transition remains a violation even if the endpoint later recovers. The comparison distinguishes an aggregate criterion from constraints on individual components.

S7 The one-stock comparator

We include this one-stock formulation as a structural comparator, not as a reduction of the two-land model. Its state A has comparator-specific stock units and is distinct from two-land state A_c. The equations are

$$
\begin{aligned}
&G(A) = \rho A\left(1 - \frac{A}{A_{\max}}\right), \qquad B(t) = b(t)A(t) + b_G G(A(t)), \qquad E(t) = eP(t);\\
&\dot{A}(t) = G\big(A(t-\tau_g)\big) - \frac{[E(t) - \sigma b(t)A(t)]_+}{b_G};\\
&\dot{P}(t) = rP(t)\left(1 - \frac{P(t)}{K(t-\tau_p)}\right), \qquad K(t) = \max\left(\frac{B(t)}{e}, K_{\min}\right);\\
&\dot{D}(t) = [E(t) - B(t)]_+ - \eta D(t), \qquad b(t) = \big(b_0 + T_b(t)\big)e^{-\alpha D(t)};\\
&T_b(t) = \omega_{Tb}\frac{\Delta b}{1 + \exp[-\kappa(t - t_w)]}.
\end{aligned}
$$

Here σ is the dimensionless human-available share; $\omega_{Tb} \in \{0, 1\}$ switches the logistic technology increment on or off. The states A, $A_{\max}$, and imposed floor A_{ext} use comparator-specific stock units. $A_{\max}$ sets the logistic growth scale; it is not an imposed upper bound. Accordingly, $G(A)$ is in stock units per year. The coefficients b, b_0, T_b, and Δb are in gha per stock unit, while b_G is in gha yr per stock unit, making both terms in B annual account values in gha.

B and E use the GFN/NFA convention, with the annual basis implicit. Population P, carrying capacity K, and $K_{\min}$ are in cap; e is in gha cap^{-1}. Debt D is in gha yr, so $\dot{D}$ and ηD are in gha, and α is in gha^{-1} yr^{-1}. Rates ρ, r, η, κ are in yr^{-1}; delays τ_g, τ_p and wave midpoint t_w are in years.

Two distinct observation operators. The comparator separates immediately available fast output σbA from the broader reported account $B = bA + b_G G(A)$. Its conversion operator responds to the unmet fast-output gap $[E - \sigma bA]_+$; the growth-linked term is valued in the aggregate but is not an additional immediately available fast output in that operator. Debt instead accumulates only the total-account gap $[E - B]_+$. Hence, even at $\sigma = 1$, conversion can be active while $E \le B$; for $A < A_{\max}$ an equilibrium can balance positive regeneration against positive depletion with zero debt input. This is a structural convention of the comparator, not an empirical allocation rule. It differs from baseline two-land conversion driven by $[E - B]_+$, so their conversion thresholds and recovery paths are not interchangeable. When $A > A_{\max}$, logistic G is negative, and neither its growth-linked account term nor this equilibrium interpretation should be described as a positive provisioning flow.

The numerical implementation enforces $A \ge A_{\text{ext}}$, $P, D \ge 0$, and $K \ge K_{\min}$, with the exact positive-part operator. Delayed histories are constant at their initial values, and RK4 integration evaluates delayed stock and account histories separately at each RK substage, using linear interpolation; the reported account always uses current $G(A)$, not the delayed regeneration input. For a zero delay the corresponding stage state is used directly. Step sizes are 0.1 yr for the full trajectory (checked at 0.05 and 0.025 yr), 0.01 yr for the fixed-demand mask (with a 0.02 yr shortfall sweep), 0.2 yr for the delay pair (checked at 0.1 yr), and 0.5 yr for the finite-grid classification.

With fixed b, $T_b = 0$, $D = 0$, and $\sigma = 1$, interior equilibria satisfying $0 < A < A_{\max}$ and $B(A)/e > K_{\min}$ obey $E = B(A)$ and $P = B(A)/e$. They form a one-parameter family along

$$
B(A) = (b + b_G\rho)A - \frac{b_G\rho}{A_{\max}}A^2, \qquad B'(A) = b + b_G\rho - 2b_G\rho\frac{A}{A_{\max}}.
$$

An interior maximum within $0 < A < A_{\max}$ requires $b_G\rho > b$. For the debt-free, constant-parameter delay and basin illustrations, $b = 0.5$, $b_G = 0.8$, and $\rho = 0.05$, so $b_G\rho = 0.04 < b$ and $B(A)$ increases throughout that interval. In the full-trajectory illustration, $b(t)$ changes with the technology wave and debt; we make no monotonicity or equilibrium-fold claim for that time-varying case.

For the debt-free reference equilibrium, $A^* = 0.8$ under $(b, b_G, \rho, A_{\max}, e, r) = (0.5, 0.8, 0.05, 1.2, 0.55, 0.02)$. It has $B^* = 0.4107\,\text{gha}$ and $P^* = B^*/e = 0.7467\,\text{cap}$. The characteristic equation has a neutral root tangent to the equilibrium family and a positive real root. That positive root is $0.5883\,\text{yr}^{-1}$ when $(\tau_g, \tau_p) = (0, 0)$ and $0.6250\,\text{yr}^{-1}$ when $(30, 25)\,\text{yr}$. These local results apply only to the comparator; they are neither a Hopf classification nor a stability result for the two-land system.

Figures S1–S4 show conditional runs of the comparator. Full-trajectory floor-contact times are 720.800, 720.750, and 720.725 yr at $\Delta t = 0.1, 0.05, 0.025\,\text{yr}$; peak-debt values are 21.8074, 21.6377, and 21.5539 gha yr, respectively. The floor-contact and debt extrema remain timestep-sensitive, so more digits are numerical diagnostics, not ecological precision. The 0.5 yr classification grid has 83 high and 125 low endpoints without regeneration delay, and 11 high and 197 low endpoints at $\tau_g = 30\,\text{yr}$, out of 208 cells each; no cells are intermediate under the stated cutoffs. In Fig. S3, the same starting state approaches $A_{\max}$ at $\tau_g = 10\,\text{yr}$ but reaches the imposed floor at $\tau_g = 30\,\text{yr}$, with $\tau_p = 25\,\text{yr}$ in both cases. This pair shows delay sensitivity for one initial condition; it does not establish a universal critical delay. Figure S4 classifies a finite grid of starting states at fixed τ_p. Grid-cell proportions are not probabilities over initial conditions or measures of a continuous basin.

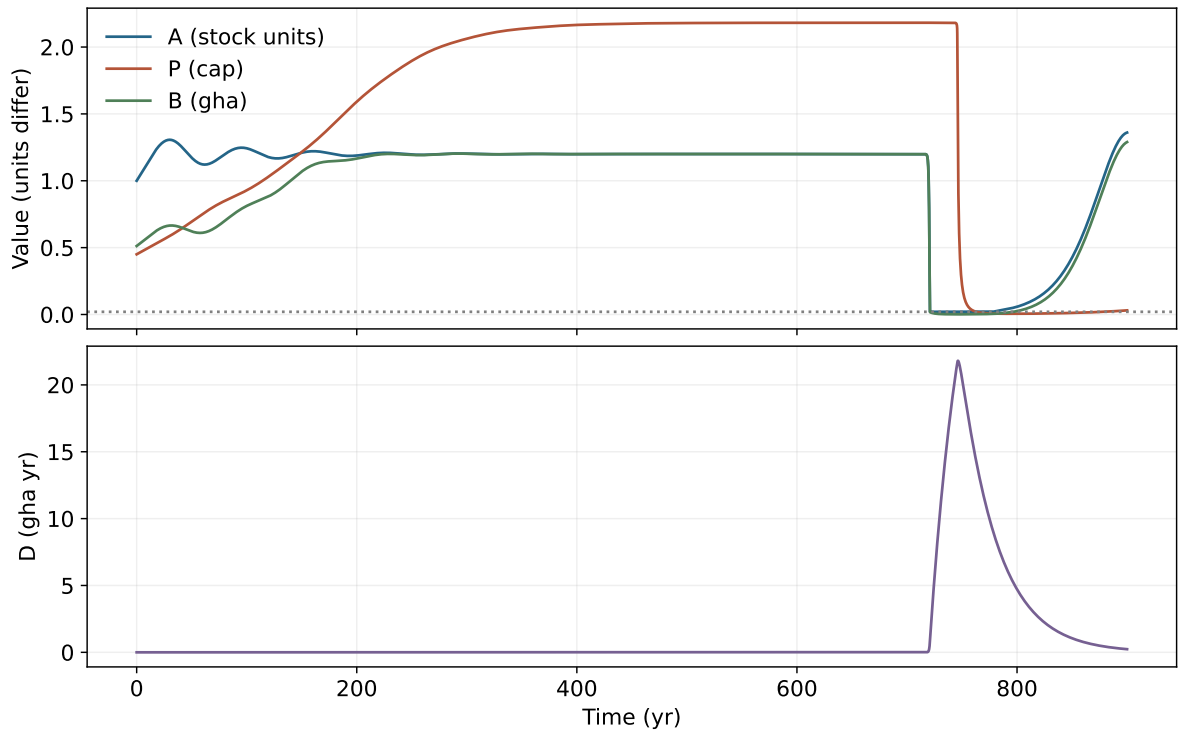


Figure S1: Full delayed one-stock comparator with a logistic technology increment. Parameters are $A_0 = 1.0$ stock unit, $P_0 = 0.45\,\text{cap}$, $b_0 = 0.5$, $b_G = 0.6$, $\rho = 0.08\,\text{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\text{gha cap}^{-1}$, $r = 0.02\,\text{yr}^{-1}$, $\eta = 0.03\,\text{yr}^{-1}$, $\alpha = 0.2\,\text{gha}^{-1}\,\text{yr}^{-1}$, $\tau_g = 15\,\text{yr}$, $\tau_p = 25\,\text{yr}$, $A_{\text{ext}} = 0.02$ stock units, $K_{\min} = 10^{-4}\,\text{cap}$, $\Delta b = 0.5\,\text{gha}$ per stock unit, $\kappa = 0.04\,\text{yr}^{-1}$, and $t_w = 120\,\text{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. The positive-part operator is exact. B and E are in gha, A is in comparator-specific stock units, P is in cap, and D is in gha yr. The trajectory reaches the imposed lower bound near 721 yr, then recovers as modeled demand falls; the bound is not absorbing. This is a conditional model illustration, not an observation or a two-land result.

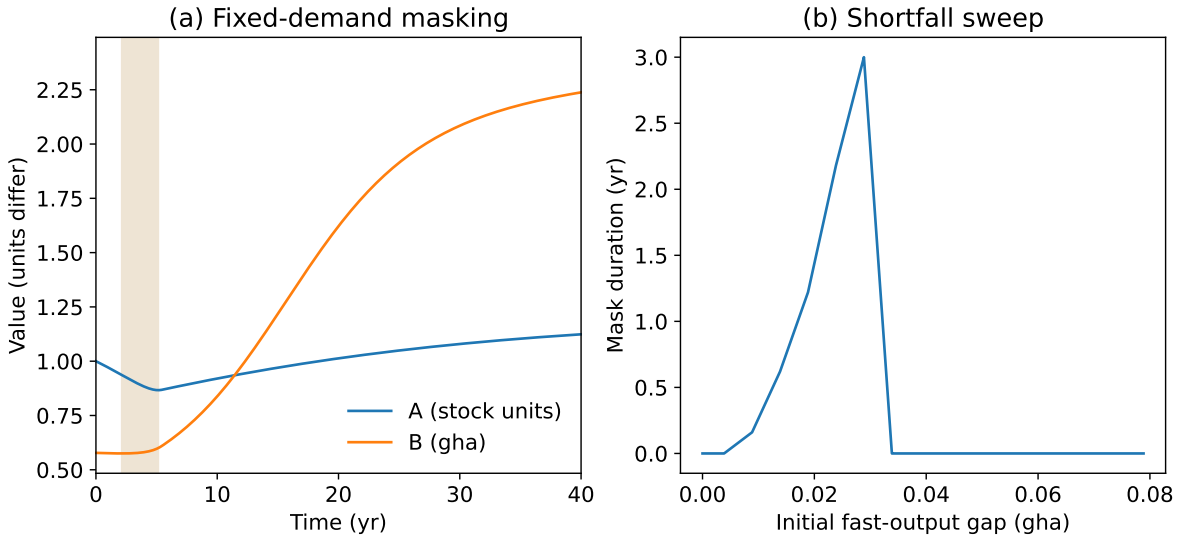


Figure S2: One-stock masking example with fixed demand and no delays. Parameters are $A_0 = 1.0$ stock unit, $E = 0.60\,\mathrm{gha}$, $b_0 = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $\eta = 0.05\,\mathrm{yr}^{-1}$, $\alpha = 0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\Delta b = 1.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.2\,\mathrm{yr}^{-1}$, and $t_w = 15\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. With the exact positive-part operator, the interval with $\dot{B} > 0$ and $\dot{A} < 0$ lasts about $3.0\,\mathrm{yr}$. Over that interval, A falls from about 0.937 to 0.867 stock units while B rises from about 0.575 to 0.600 gha. Panel (b) shows the interval width across initial shortfalls $[E(0) - b(0)A_0]_+$ for this parameterization. These are conditional comparator outputs.

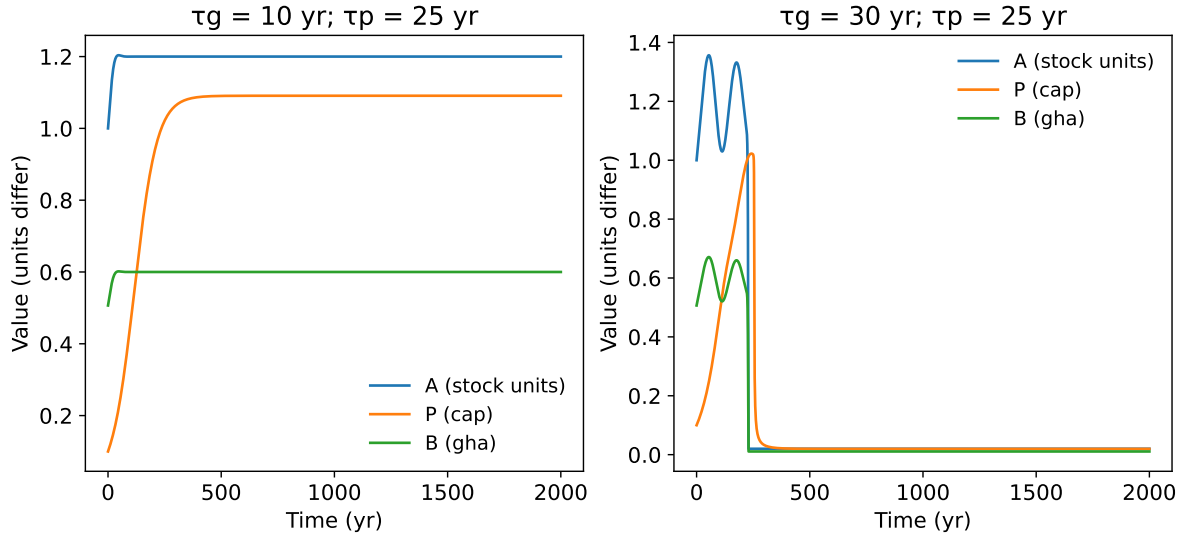


Figure S3: Controlled regeneration-delay examples in the debt-free one-stock subsystem. Both runs use $A_0 = 1.0$ stock unit, $P_0 = 0.1\,\text{cap}$, $b = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\text{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\text{gha}\,\text{cap}^{-1}$, $r = 0.02\,\text{yr}^{-1}$, $\tau_p = 25\,\text{yr}$, $A_{\text{ext}} = 0.02$ stock units, $K_{\min} = 10^{-6}\,\text{cap}$, and $\sigma = 1$; only τ_g differs (10 versus 30 yr). B is in gha, P in cap, and A in comparator-specific stock units. These selected trajectories do not establish a general critical delay.

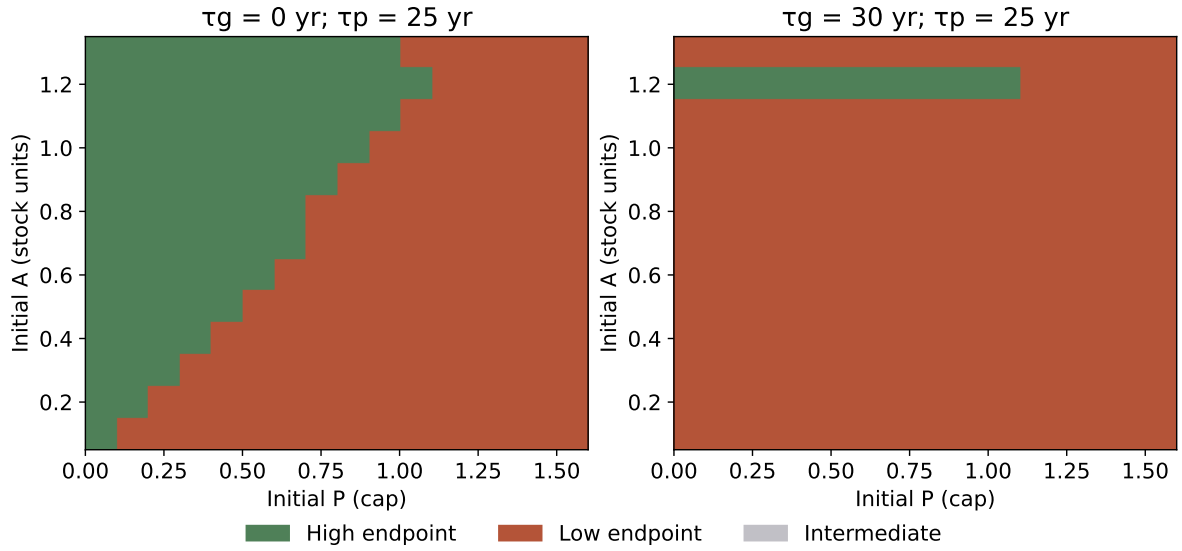


Figure S4: Finite-grid initial-condition classification in the debt-free one-stock subsystem. The runs use the parameter set in Fig. S3, with $K_{\min} = 10^{-6}\,\text{cap}$ and $\tau_p = 25\,\text{yr}$; $\tau_g = 0\,\text{yr}$ in the left panel and 30 yr in the right. The imposed lower bound is $A_{\text{ext}} = 0.02$ stock units. The grid uses $A_0 = 0.10, 0.20, \ldots, 1.30$ stock units and $P_0 = 0.05, 0.15, \ldots, 1.55\,\text{cap}$; trajectories run to 1200 yr with $\Delta t = 0.5\,\text{yr}$. Green marks a high endpoint $A(1200) \geq 0.95 A_{\max}$, red a low endpoint $A(1200) \leq 0.05$ stock units, and gray an intermediate endpoint. These display-grid categories are not probabilities or continuous-basin measures.

S8 Controlled technology-wave experiment

The matched experiment compares endogenous-demand runs with the technology increment T_b switched on or off, at $P_0 = 1.0$ or $1.2\,\mathrm{cap}$. Within each pair, the runs share $A_{tot} = 2.5\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $A_{c0} = 0.9\,\mathrm{ha}$, $q_0 = 0.9$, and $D_0 = 0.0\,\mathrm{gha\,yr}$. Parameters and numerical methods are matched; only ω_{Tb} differs. Both runs also share constant pre-run histories at $q_0 = 0.9$ and the no-wave initial account value $B = 0.560184\,\mathrm{gha}$.

Shared parameter values are $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$, $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$, $b_{G,c} = 0.8\,\mathrm{gha\,yr\,ha^{-1}}$, $\rho_c = 0.08\,\mathrm{yr^{-1}}$, $\alpha = 0.03\,\mathrm{gha^{-1}\,yr^{-1}}$, $\eta = 0.05\,\mathrm{yr^{-1}}$, $\eta_f = 0.05\,\mathrm{yr^{-1}}$, $\mu = 0.06\,\mathrm{yr^{-1}}$, $e = 0.55\,\mathrm{gha\,cap^{-1}}$, $r = 0.02\,\mathrm{yr^{-1}}$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $\tau_{conv} = 1\,\mathrm{yr}$, and $\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$. The active wave uses $\Delta b = 1.0\,\mathrm{gha\,ha^{-1}}$, $\kappa_w = 0.1\,\mathrm{yr^{-1}}$, and $t_{wave} = 40\,\mathrm{yr}$; runs last $250\,\mathrm{yr}$. The logistic increment is already nonzero at $t = 0$ ($T_b(0) = 0.017986\,\mathrm{gha\,ha^{-1}}$), so initial B is $0.560184\,\mathrm{gha}$ without the wave and $0.570976\,\mathrm{gha}$ with it. This small difference comes from the logistic tail. Land, quality, population, and debt states otherwise match, and the pre-run B history remains common. The wave reaches half its increment at $t_{wave} = 40\,\mathrm{yr}$.

Figure S5 shows the $P_0 = 1.0$ pair. In both runs, peak debt occurs after the wave midpoint: $D_{max} = 0.008126\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ without the wave and $0.020205\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ with it. At $P_0 = 1.2$, the all-time maximum occurs early and is lower with the wave, but the maximum for $t \geq t_{wave}$ is higher. Table S13 reports both windows. Because t_{wave} marks the logistic midpoint, the post-midpoint window includes the wave's continued rise.

Table S13: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250\,\mathrm{yr}$; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

For $P_0 = 1.0$, the all-time wave/no-wave peak-debt ratio is 2.487 at $\Delta t = 0.05\,\mathrm{yr}$ and 2.492 at $0.025\,\mathrm{yr}$; under current-quality regeneration it is 2.422 at $0.05\,\mathrm{yr}$. For $P_0 = 1.2$, all-time ratios are 0.834 and 0.833, while post-midpoint ratios are 2.320 and 2.328. Halving the step changes debt maxima by less than 0.9%. In the $P_0 = 1.0$ pair, A_c stays at or above its initial $0.9\,\mathrm{ha}$, so this experiment illustrates a conditional delayed debt rebound, not the fixed-demand capital-land drawdown. At $P_0 = 1.2$, the all-time and post-midpoint maxima tell different stories. The debt variable D is a leaky state driven by aggregate deficit $E - B$, not an observed NFA or physical-stock quantity.

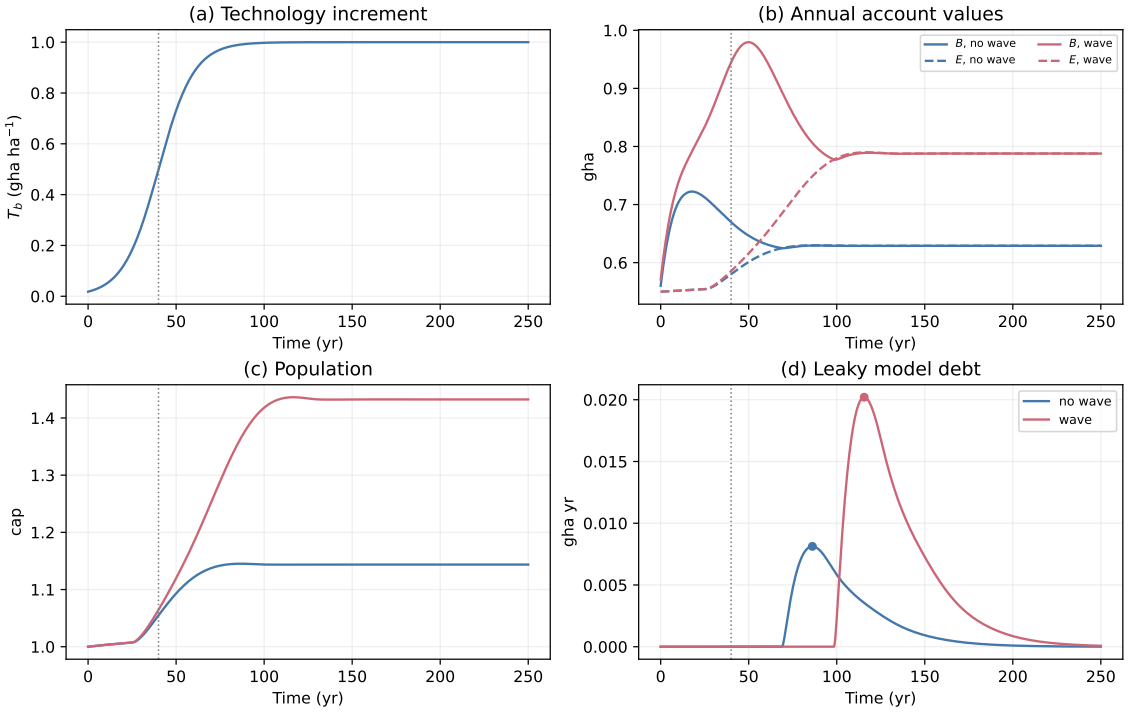


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); and (d) leaky model debt D (gha yr). Markers denote all-time maxima: 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}. The dotted line marks the midpoint, not an end date.

S9 Sharp static identification bounds and witnesses

The intervals in main-text Proposition 4 follow from the static observation relation and the stated allocation restrictions. The independently known inputs are total area $L = A_{tot}$, minimum capital area $a_{\min}$, current fast-land yield b_f, and a continuously attainable, area-independent interval $[u_-, u_+]$ for $u = \Upsilon_c(q)$, with $0 < u_- \le u_+ < b_f$. All land is nonnegative and $A_c \ge a_{\min}$. Additional joint restrictions on yields and land allocations require projection of the corresponding joint feasible set.

S9.1 Proof and feasibility construction

Fix $A_c = a \in [a_{\min}, L]$. The remaining choices are $A_f \in [0, L-a]$ and $u \in [u_-, u_+]$, with $A_r = L - a - A_f$. The continuous affine map $b_f A_f + ua$ maps this connected rectangle onto the full interval

$$[\, u_- a,\ b_f(L-a) + u_+ a \,].$$

Thus a compatible allocation exists exactly when $u_- a \le B \le b_f(L-a) + u_+ a$. Since $b_f - u_+ > 0$, solving gives

$$a_{\min} \le a \le \min\{L, B/u_-, (b_f L - B)/(b_f - u_+)\}.$$

This is a set equality, not merely an outer bound. For an explicit construction, choose any u in

$$[u_-, u_+] \ \cap\ [(B - b_f(L-a))/a,\ B/a]$$

when $a > 0$, and set $A_f = (B - ua)/b_f$, $A_r = L - a - A_f$. The inequalities guarantee the intersection is nonempty. At $a = 0$, feasibility instead requires $0 \le B \le b_f L$, and $A_f = B/b_f$ supplies the construction.

When A_f is independently observed, the residual $C^{\mathrm{mod}} = B - b_f A_f$ obeys $C^{\mathrm{mod}} = ua$. For $C^{\mathrm{mod}} \ge 0$, intersecting $C^{\mathrm{mod}}/u_+ \le a \le C^{\mathrm{mod}}/u_-$ with $[a_{\min}, L - A_f]$ gives the second interval. Choosing $u = C^{\mathrm{mod}}/a$ constructs every feasible positive a; when $a = C^{\mathrm{mod}} = 0$, any permitted u works. If $C^{\mathrm{mod}} < 0$, the residual is inconsistent with nonnegative capital area and positive yield. If q is also independently observed, its known $\Upsilon_c(q) > 0$ gives the unique feasible candidate $a = C^{\mathrm{mod}}/\Upsilon_c(q)$.

Attainment of each u by an admissible quality value is essential to calling these intervals sharp. Under the baseline law $\Upsilon_c(q) = 0.05 + 0.064q(1-q)$ on $[0,1]$, the yield image is exactly $[0.05, 0.066]\,\mathrm{gha\,ha^{-1}}$. Its maximum occurs at $q = 0.5$; its minimum at $q = 0$ or 1. This is a readout property, not a monotonic ordering of ecological condition. In particular, $q = 0.1$ and 0.9 give the same Υ_c, so known Υ_c alone does not uniquely identify q.

S9.2 Numerical illustration and interpretation

For $B = 0.825\,\mathrm{gha}$, $b_f = 0.85\,\mathrm{gha\,ha^{-1}}$, $L = 2.5\,\mathrm{ha}$, $a_{\min} = 0.05\,\mathrm{ha}$, and the above yield interval, the B-only interval is $[0.05, 1.6581632653\ldots]\,\mathrm{ha}$. Independently fixing $A_f = 0.785/0.85\,\mathrm{ha}$ gives $C^{\mathrm{mod}} = 0.04\,\mathrm{gha}$ and $A_c \in [0.6060606061\ldots, 0.8]\,\mathrm{ha}$. Table S14 supplies endpoint witnesses. They satisfy the identical B and A_f constraints and respect total area; neither is asserted to be reachable from the baseline history. The lower endpoint of the wider B-only interval is feasible, for example, with $a = 0.05$, $u = 0.05$, and $A_f = (0.825 - 0.0025)/0.85\,\mathrm{ha}$. Its upper endpoint is feasible at $u = 0.066$, $A_r = 0$.

At known $q = 0.9$, $A_c = 0.04/0.05576 = 0.7173601148\ldots\,\mathrm{ha}$. The illustrative reporting cut of 0.7 ha lies inside the interval identified from B and independently known A_f, whereas knowing q identifies a value above that cut. The cut illustrates set overlap rather than an ecological threshold or confidence level. These deterministic bounds assume exact inputs and a valid observation operator. With uncertain B, b_f, A_f or yield restrictions, define a coherent joint feasible uncertainty set and project it onto A_c; independent endpoint combinations can otherwise

Table S14: Endpoint witnesses for the B, A_f interval. Areas are in ha, u in gha ha^{-1}, B in gha, and q is dimensionless. Values shown are rounded; the calculations use unrounded A_f.

Endpoint	A_f	A_c	A_r	q	u	Calculated B
lower	0.923529	0.606061	0.970410	0.5	0.066	0.825
upper	0.923529	0.800000	0.776471	1.0	0.050	0.825

be inconsistent. No empirical mapping from NFA land accounts to these physical model areas is established here. The construction narrows an identified set; it does not establish dynamic observability or make unobserved quality measurable.

S10 Integrated contribution budgets and numerical refinement

S10.1 Continuous-time identity and quadrature

Let $u = \Upsilon_c(q)$ and $M = \mu A_r - \eta_f A_f$. Differentiating the observation relation and substituting the land equations gives, almost everywhere along the absolutely continuous projected trajectory,

$$\dot{B} = (b_f - u)u_c + b_f M + uR_{rc} + b_{G,c}G_c'(q)A_c\dot{q} + A_f\dot{b}_f, \qquad \dot{b}_f = e^{-\alpha D}\dot{T}_b - \alpha b_f \dot{D}.$$

The projected quality velocity is used in this identity. Each integrand has units gha yr^{-1} and each time integral has units gha. The fast-yield contribution comprises $A_f e^{-\alpha D}\dot{T}_b$ from technology and $-\alpha A_f b_f \dot{D}$ from debt. Debt accumulation reduces fast-land yield, whereas debt decay increases it. Their integrated contributions summarize the signed effect over the observation interval; peak debt is a separate statistic.

All five continuous-time integrands are evaluated at the numerical state nodes and integrated using the trapezoidal rule. The endpoint difference is evaluated separately from the observation relation. The signed closure residual is

$$\varepsilon_T = B(T) - B(0) - \sum_j I_j,$$

where I_j denotes an integrated contribution. The integrands are evaluated directly from the continuous right-hand side. This quadrature differs from a telescoping discrete product decomposition, which closes by construction. Finite-step discrepancies reflect the Euler trajectory approximation, quadrature error and the non-smooth quality constraint. Timestep refinement assesses the resulting closure error; it does not provide a uniform trajectory-error bound.

S10.2 Simulation conditions

Four conditions are considered. The fixed-demand condition uses $A_{c0} = 1.5\,\mathrm{ha}$, $P_0 = 1.5\,\mathrm{cap}$, $E = 0.825\,\mathrm{gha}$, $\chi_r = 0$ and $T = 100\,\mathrm{yr}$. The restoration condition uses $A_{c0} = 1.0\,\mathrm{ha}$, $P_0 = 2.0\,\mathrm{cap}$, $E = eP$, $\chi_r = 0.1\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$ and $T = 400\,\mathrm{yr}$. The control and technology conditions use $A_{c0} = 0.9\,\mathrm{ha}$, $P_0 = 1.0\,\mathrm{cap}$, endogenous demand, $\chi_r = 0.1\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$ and $T = 250\,\mathrm{yr}$; only the technology condition includes the logistic yield increment specified in S8. All conditions share $A_{f0} = 0.6\,\mathrm{ha}$, $q_0 = 0.9$, $D_0 = 0$, $A_{tot} = 2.5\,\mathrm{ha}$ and the baseline coefficients and delays. Population evolves in the fixed-demand condition, but demand remains exogenous.

At each Euler step, the current state is evaluated at the preceding node, while the technology input and delayed observations are evaluated at the right endpoint. Current supply is calculated on the preceding state using that technology input; post-update supply is stored for subsequent delayed interpolation. The pre-run quality history is constant at 0.9. Pre-run supply is the no-technology initial value, including in the matched technology comparison. The quality velocity is projected before updating, followed by clipping to the admissible interval. Nonnegative bounds are imposed on A_f, P, D, and $A_c \geq 0.05\,\mathrm{ha}$. Reserve area is calculated from total-area conservation and monitored. The optional fast-land ceiling is inactive.

At $\Delta t = 0.05\,\mathrm{yr}$, peak debt is $0.0081259288\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ in the control condition and $0.0202046739\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ in the technology condition (Figure S5). The contribution integrals use the same mathematical model and history specifications as these simulations.

S10.3 Contribution integrals and closure

Table S15 reports integrated contributions at the smallest timestep. Table S16 separates technology and debt within the fast-yield contribution. The positive technology contribution is substantially offset by the negative maintenance–retirement contribution along the technology trajectory. These are decompositions of coupled model trajectories, not estimates of empirical causal effects.

Table S15: Integrated contributions to model biocapacity in gha at $\Delta t = 0.00078125$ yr. Horizons are 100 yr for fixed demand, 400 yr for restoration, and 250 yr for the control and technology conditions. Residuals are the endpoint change minus the sum of integrated contributions.

Contribution or statistic	Fixed demand	Restoration	Control	Technology
Conversion	+0.547925	+0.513933	+0.016593	+0.042011
Maintenance/retirement	-0.311098	-0.218037	+0.037254	-0.487030
Restoration	+0.000000	+0.016561	+0.020431	+0.046986
Quality readout	-0.006470	-0.002577	-0.005498	-0.005567
Fast-yield change	+0.000873	+0.002687	+0.000001	+0.620615
Sum of integrals	+0.231230	+0.312567	+0.068781	+0.217016
$B(T) - B(0)$	+0.231310	+0.312764	+0.068791	+0.217025
Closure residual	+0.000080	+0.000197	+0.000010	+0.000008

Table S16: Technology and debt components of the fast-yield integral, in gha, at the smallest timestep. A small signed debt integral is compatible with nonzero debt at intermediate times.

Condition	Technology	Debt	Fast-yield total
Fixed demand	+0.00000000	+0.00087257	+0.00087257
Restoration	+0.00000000	+0.00268738	+0.00268738
Control	+0.00000000	+0.00000057	+0.00000057
Technology	+0.62061305	+0.00000241	+0.62061546

The direct restoration contribution to B is distinct from restoration's total effect on the evolving state.

Seven nested timesteps are used: 0.05, 0.025, 0.0125, 0.00625, 0.003125, 0.0015625 and 0.00078125 yr. At the smallest timestep, absolute closure residuals relative to the endpoint changes are 0.0348%, 0.0628%, 0.0143% and 0.00380% for fixed demand, restoration, control and technology, respectively. These satisfy the closure tolerance $|\varepsilon_T|/|\Delta B| < 0.001$. Absolute residuals decrease at every tested timestep halving in all four conditions (Table S17). Maximum differences between nested-grid solutions for A_c, B, D are provided with the computational data.

Table S17: Timestep refinement of the integrated contribution budget. ΔB, $\sum I_j$ and $\varepsilon_T = \Delta B - \sum I_j$ are in gha; timestep is in yr.

Condition	Δt	ΔB	$\sum I_j$	ε_T
Fixed demand	0.05000000	+0.23131065	+0.22616616	+5.144e-03
Fixed demand	0.02500000	+0.23131037	+0.22873531	+2.575e-03
Fixed demand	0.01250000	+0.23131023	+0.23002006	+1.290e-03
Fixed demand	0.00625000	+0.23131016	+0.23066538	+6.448e-04
Fixed demand	0.00312500	+0.23131012	+0.23098805	+3.221e-04
Fixed demand	0.00156250	+0.23131010	+0.23114939	+1.607e-04
Fixed demand	0.00078125	+0.23131009	+0.23122970	+8.040e-05
Restoration	0.05000000	+0.31300381	+0.30043090	+1.257e-02
Restoration	0.02500000	+0.31288178	+0.30659261	+6.289e-03
Restoration	0.01250000	+0.31282082	+0.30967470	+3.146e-03
Restoration	0.00625000	+0.31279032	+0.31121731	+1.573e-03
Restoration	0.00312500	+0.31277507	+0.31198869	+7.864e-04
Restoration	0.00156250	+0.31276745	+0.31237439	+3.931e-04
Restoration	0.00078125	+0.31276363	+0.31256709	+1.965e-04
Control	0.05000000	+0.06881854	+0.06818752	+6.310e-04
Control	0.02500000	+0.06880431	+0.06848617	+3.181e-04
Control	0.01250000	+0.06879720	+0.06863550	+1.617e-04
Control	0.00625000	+0.06879364	+0.06871312	+8.052e-05
Control	0.00312500	+0.06879187	+0.06875193	+3.993e-05
Control	0.00156250	+0.06879098	+0.06877134	+1.964e-05
Control	0.00078125	+0.06879053	+0.06878067	+9.860e-06
Technology	0.05000000	+0.21685345	+0.21632588	+5.276e-04
Technology	0.02500000	+0.21694039	+0.21667384	+2.666e-04
Technology	0.01250000	+0.21698384	+0.21684782	+1.360e-04
Technology	0.00625000	+0.21700556	+0.21693788	+6.768e-05
Technology	0.00312500	+0.21701642	+0.21698291	+3.351e-05
Technology	0.00156250	+0.21702185	+0.21700543	+1.642e-05
Technology	0.00078125	+0.21702456	+0.21701631	+8.251e-06

The numerical bounds on A_f, A_c, P, D require no clipping in these 28 simulations, and the conversion-availability constraint is inactive. Each simulation contains one finite-step overshoot of the quality ceiling followed by clipping. At the smallest timestep, the maximum pre-clipping overshoot is 1.25×10^{-6}, and the first constrained node occurs at 13.8890625 yr. Minimum reserve areas are 0.400000, 0.775769, 0.531279 and 0.286300 ha for fixed demand, restoration, control and technology, respectively; all exceed the monitored 0.1 ha threshold. Total area is conserved to floating-point precision.

For the restoration condition at $\Delta t = 0.00078125$ yr, $\min A_c = 0.401217$ ha and $A_c(400) = 0.678295$ ha, indicating partial recovery. The $\Delta t = 0.1$ yr estimates are given in Table S5. Quality trajectories for the specified 20 and 40 yr delays coincide because the quality constraint is reached before either delay elapses (S2.3 and S3). The budget calculations therefore provide no additional inference about sensitivity to alternative quality histories.

The computational data contain full-precision contribution integrals, signed closure residuals, state-constraint statistics and nested-grid comparisons. Trajectories and cumulative integrals are reported at 0.05 yr intervals; integration uses every simulation node.

S10.4 Executable data and observation specifications

The companion package provides an offline implementation: `reproduce_empirical_C18_A01.py` generates annual observations, endpoint and component tables, proxy sensitivities and main empirical figures; `reproduce_two_land_C18_A01.py` generates the drawdown, recovery, threshold,

and gate-sign outputs; and `reproduce_comparator_C18_A01.py` generates the one-stock trajectories, finite-grid classifications, characteristic roots and supplementary figures. Input hashes, release metadata, source URLs and attribution are supplied with the data. All scripts run relative to their own package root, without a network request. The current-quality sensitivity is named separately from the selected delayed-quality specification.

S11 Operational prediction and falsification map

S11.1 Independent observations and test separation

The tests below distinguish three objects: an accounting identity, a conditional dynamical prediction, and the validity of a model–observation mapping. An identity computed from the same account inputs is not independent validation. In particular, NFA total and component entries can test arithmetic reproduction, but cannot alone identify A_c, q, conversion, or restoration. The empirical remainder $C^{\mathrm{emp}} = B - X$ is never substituted for $C^{\mathrm{mod}} = \Upsilon_c(q)A_c$ in these tests.

A physical application must first delimit a spatial system with non-overlapping land classes, record an independent land-transition matrix, and measure fast-land output per area and capital-area condition using a stated observation operator. Condition indicators such as biomass density or soil carbon need an externally justified mapping to q; neither indicator is assumed to be this dimensionless state. Account conversion coefficients must be fixed or estimated on calibration observations distinct from the validation interval. A failed mapping is a substantive rejection of that physical application, even when the model's internal algebra remains correct.

Table S18: Operational tests of P1–P5. Each rejection criterion is conditional on the independently checked assumptions in its row. No row is declared empirically passed by the present world-account decomposition.

Item	Independent data and quantities to estimate	Estimand and rejection outcome
P1	Measured converted area v, receiving and donor per-area annual contributions under a common valuation, and independent maintenance, retirement, restoration, condition and technology changes. Estimate b_f and $\Upsilon_c(q)$ without using the held-out total B to define either contribution. Prefer randomized or otherwise identified matched conversion interventions.	The conversion-only account change per area is $b_f - \Upsilon_c(q)$. A persistently opposite sign, or a held-out discrepancy exceeding propagated measurement and counterfactual uncertainty, rejects the stated mapping/mechanism after other flows are accounted for. Rising net B by itself neither confirms nor rejects this marginal prediction.
P2	Local, matched perturbations of demand or area, repeated across operating states, with fast/capital/reserve areas and demand observed. Estimate μ, η_f, τ_{conv}, and yield coefficients on calibration intervals; maintain the specified active-shortfall branch, fixed quality/debt and no restoration for the reduced test.	Estimate the dominant relaxation rate from the area response and compare with the predicted slow eigenvalue. Reproducible response times outside predictive uncertainty reject the calibrated reduced-rate prediction. A delay, switched regime, or changing coefficients must not be silently fitted as the same local experiment.

Item	Independent data and quantities to estimate	Estimand and rejection outcome
P3	A sequence of controlled or identified demand levels approaching the calibrated reduced ceiling, with repeated perturbation responses and capital-area minima. Independently check the fixed-parameter, debt-free, cap-quality, no-restoration assumptions and the one-sided active-shortfall regime.	The predicted deficit-side matrix has a demand-independent spectrum, and the equality relation reaches the area floor without a fold. An estimated relaxation rate tending to zero or a reproducible turning branch under those restrictions rejects this reduced model application. Outside those restrictions, such behavior tests a different system; finite drawdown samples cannot prove general continuity.
P4	For software verification, supply the exact constant history $q(s) = 0.9$, $\rho_c = 0.08$, the imposed cap 1, and delays 20 or 40 yr. For a physical application, independently measure condition and identify its mapping, history, cap and regeneration rate rather than estimating them from equality of two simulated runs.	The specified projected model gives $q(t) = \min(1, 0.9 + 0.0072t)$ and contact at 125/9 yr. A nonvanishing integration discrepancy under refinement rejects its implementation. A mapped physical path incompatible with this law rejects the cap/history model for that system. Equality of the two constructed paths is not an empirical claim of general delay insensitivity.
P5	Observe capital-area losses and returns from a transition ledger; distinguish restoration from natural succession, reclassification and migration of boundaries. Compare demand reduction alone with matched return-flow interventions. Estimate conversion and return rates independently, and record the sign of $B - E$ when the policy gate acts.	With $R_{rc} = 0$, the model predicts $\dot{A}_c = -u_c \leq 0$: a verified positive area return with no modeled return flow rejects the closed land-flow specification for that application. Under the declared surplus gate, positive restoration during $B < E$ rejects the gate mapping. A lower demand or later higher B is not itself evidence of restoration.

S11.2 Sampling, uncertainty and prospective rejection rules

The validation unit, spatial boundary, calibration/validation split, intervention definition, and mapping from physical observations to model variables must be declared before inspecting validation outcomes. Measurements should resolve the fastest tested response: as a design target, spacing no larger than one-fifth of the shortest relevant conversion, demographic or regeneration timescale avoids treating an unresolved transient as a measured relaxation rate. Observation windows must also cover the delayed inputs and several predicted return times. The illustrative 1 yr conversion constant would require substantially finer observations than the annual global account series. This is a design requirement, not a claim that the present data meet it.

Parameter and measurement uncertainty should be propagated jointly, retaining covariance between area, yield and valuation. A prospective protocol can use a predeclared 95% predictive interval for its primary held-out estimand; repeated tests require a prespecified family-wise or false-discovery adjustment. Serial observations require a dependence-aware interval or block re-

sampling with block length chosen on calibration data. Exceedance must be replicated across the predeclared independent intervention units rather than counted repeatedly along one autocorrelated trajectory. No numerical effect-size or power guarantee follows from these illustrative parameters; pilot measurement error and an intervention effect are needed to size a field test.

For P2–P3, regime assumptions are checked independently of whether the fitted relaxation favors the prediction; cases failing those checks are reported as out-of-scope, not silently removed to improve agreement. For P1 and P5, total-account errors and land-transition errors are reported separately. Rejection of a physical mapping cannot be repaired merely by redefining the proxy remainder to equal the model capital contribution. These requirements make the proposed empirical comparison falsifiable without treating internal consistency or account arithmetic as dynamic ecological validation.

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